

# Project 2 - Computer-assisted Mathematics

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Bonn, February 2026

## 1 Main Results

In this project, I formalized the following theorem:

**Theorem 1** *Let  $F$  be an algebraically closed field, and let  $SL(2, F)$  denote the  $2 \times 2$  matrices with coefficients in  $F$  and determinant 1. Then  $SL(2, F)$  acts triply transitive on the projective line  $L$  over  $F$ .*

To prove this theorem, we first need to define the group action of  $SL(2, F)$  on the projective line  $L$  and the notion of a group action being *triply transitive* on a set.

The projective line is defined as the quotient space of a two-dimensional  $F$  vector space  $V$ . The equivalence relation of this quotient space is given by multiplication with a nonzero scalar in  $F$ , meaning that a point on the projective line corresponds to a one-dimensional subspace of  $V$ . Therefore, it is natural to take an arbitrary representative  $p \in V$  of a point  $P \in L$  and multiply it by a  $2 \times 2$  matrix  $\Phi \in SL(2, F)$  (after choosing a basis for  $V$ ). We then define  $\Phi \bullet P$  as the projection of  $\Phi \bullet p$ . This gives a well defined group action.

**Definition 1** *A group  $S$  acts triply transitive on a set  $X$  if, for all subsets  $\{x_1, x_2, x_3\}$  and  $\{y_1, y_2, y_3\}$  with  $i \neq j \implies x_i \neq x_j \wedge y_i \neq y_j$ , there exists an element  $s \in S$  such that  $\forall i : s \bullet x_i = y_i$ .*

The idea of the proof is to take arbitrary representatives  $p_1, p_2, p_3, q_1, q_2$ , and  $q_3 \in V$  of six points  $P_1, P_2, P_3, Q_1, Q_2$ , and  $Q_3$  on the projective line  $L$ . Since all  $p_i$  are pairwise linearly independent and all  $q_i$  are pairwise linearly independent, we can construct a non-singular linear map  $\pi : V \rightarrow V$  mapping  $\alpha \cdot p_1 \mapsto \gamma \cdot q_1$ ,  $\beta \cdot p_2 \mapsto \delta \cdot q_2$ , and  $p_3 \mapsto q_3$  for some  $\alpha, \beta, \gamma, \delta \in F \setminus \{0\}$ .

Then the map  $\phi : V \rightarrow V := \pi \cdot \left( \frac{1}{\sqrt{\det(\pi)}} \right)$  has determinant 1 and maps, for all  $i \in \{1, 2, 3\}$ , a multiple of  $p_i$  to a multiple of  $q_i$ . In other words, the matrix  $\Phi \in SL(2, F)$  associated to the map  $\phi$  satisfies  $\forall i \in \{1, 2, 3\} : \Phi \bullet P_i = Q_i$ .

## 2 Sorrys

In the proof, there is one sorry that I was not able to prove. It remains to show that, given a linear automorphism  $\phi : V \rightarrow V$  and a basis  $b$  for  $V$ , the following procedure is equal to applying the map  $\phi$ : first take the  $b$ -representation of a vector  $v \in V$ , multiply it by the matrix representation of  $\phi$  with respect to the basis  $b$ , and then return from  $F^2$  to  $V$  via the chosen basis  $b$ .

I attempted to solve this using the theorem “Module.Basis.toLin.toMatrix “, which states that the linear map representing a change-of-basis matrix is the identity on the vector space  $V$ .

## 3 References and Sources

The main source I used was “*Christopher Butler - Classification Of Finite Subgroups Of The Two-Dimensional Special Linear Group Over An Algebraically Closed Field (2019)*“, pages 18 and 19. This document contains not only the theorem but also the definition of triply transitive and explains how  $SL(2, F)$  acts on the projective line.

## 4 AI

I used leansearch extensively to find useful lemmata and theorems. If I got stuck on a problem, I sometimes asked ChatGPT, which never provided a correct proof but did give me new ideas on how to proceed. For example, I did not know previously that I could prove goals about the quotient space by induction. Therefore, even though I used AI, I would not say that any part of my project was not created by myself.

I also used ChatGPT to correct linguistic errors in this text.